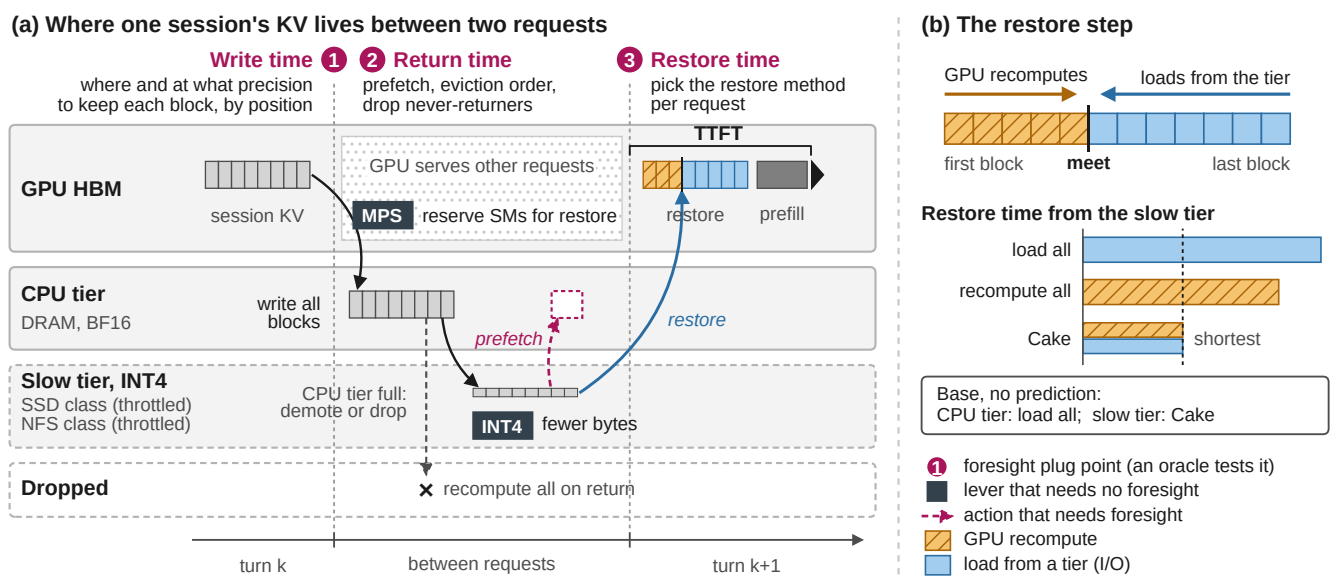

(a) Where one session's KV lives between two requests
Write time
1
where and at what precision to keep each block, by position
2
Return time
prefetch, eviction order, drop never-returners
3
Restore time
pick the restore method per request
TTFT
GPU HBM
session KV
GPU serves other requests
MPS
reserve SMs for restore
restore
prefill
CPU tier
DRAM, BF16
write all blocks
prefetch
restore
Slow tier, INT4
SSD class (throttled)
NFS class (throttled)
CPU tier full: demote or drop
INT4
fewer bytes
Dropped
recompute all on return
turn k
between requests
turn k+1
(b) The restore step
GPU recomputes
loads from the tier
first block
meet
last block
Restore time from the slow tier
load all
recompute all
Cake
shortest
Base, no prediction:
CPU tier: load all; slow tier: Cake
foresight plug point (an oracle tests it)
lever that needs no foresight
action that needs foresight
GPU recompute
load from a tier (I/O)